Product Design Specification (PDS)

1. Performance & Functional Requirements

- **Primary Function:** An interactive basketball-themed toy where a ball is lifted, sorted, and shot into a hoop.
- **Sorting Mechanism:** A sensor-based system designed to differentiate balls specifically by **color**.
- **Lifting Mechanism:** A **continuously rotating spiral screw** (Archimedes' screw) to transport balls vertically.
- **Shooting Mechanism:** A projectile system calibrated to launch the ball into a basketball hoop.
- **Closed-Loop System:** A single-entry point for marbles/balls that must remain within the internal track throughout the cycle.
- **Interactive Features:** Must include at least one embedded interactive feature (e.g., buttons or sensors) and a dedicated Start/Stop button.

2. Electronics & Control System

- **Microcontrollers:** Two microcontrollers (e.g., Arduino Uno) using a standard communication protocol.
- **User Interface 1:** An onboard **LCD/TFT Touch Display** for direct interaction (No 7-segments).
- **User Interface 2:** A remote application built in **Processing IO** communicating via **Bluetooth**.
- **Custom Electronics:** At least one custom-manufactured **PCB** must be integrated into the system.
- **Power:** Powered by a **single unified power source**.

3. Physical & Material Specifications

- **Manufacturing:** Primary structural and mechanical components will be **3D printed**.
- **Dimensions:** Total assembly must not exceed **100 cm x 100 cm**.
- **Modularity:** The design must be divided into modules with individual wiring plugs for easy assembly and disassembly.
- **Safety:** All wiring must be hidden; the 3D-printed surfaces must be smooth and safe for the target age group.

4. Target Audience

- **User Profile:** Designed for children aged **8 years old ($\pm$ 6)**, focusing on engagement levels suitable for older children while maintaining safety for younger ones.